INTRODUCTION TO INFORMATION SECURITY

WEEK 6

Transposition Ciphers

Transposition Ciphers

The Transposition Cipher Technique is an encryption method used to encrypt a message or information

- Uses systematic shuffling of plain text characters or bits to secure data by altering their positions based on some defined way or algorithm.
- The positions of the characters present in the plaintext are rearranged or shifted to form the ciphertext.

Types

- Rail Fence Transposition Cipher
- Block (Single Columnar) Transposition Cipher
- Double Columnar Transposition Cipher

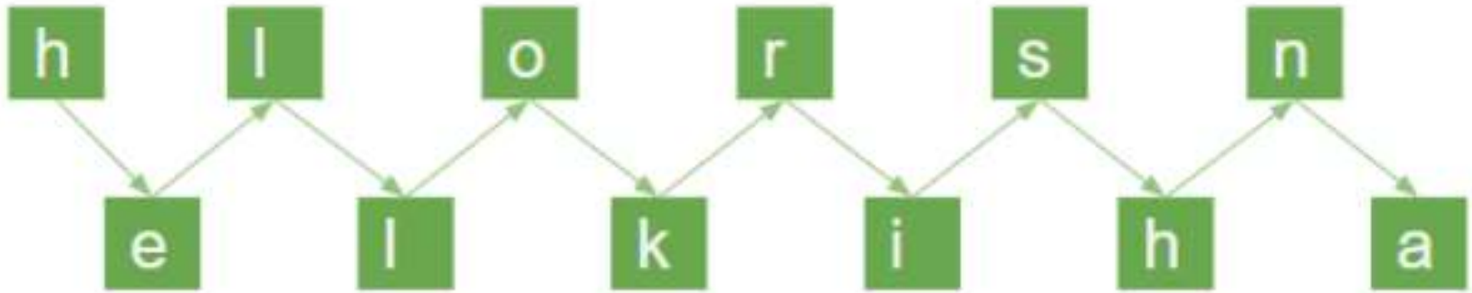
Rail Fence Transposition Cipher

- Rail Fence Transposition cipher technique is the simplest transposition cipher technique. It is also termed as a zigzag cipher.
- It gets its name from the way through which it performs encryption of plain text.

Rail Fence Transposition Cipher

Example: The plain text is "Hello Krishna"

Now, we will write this plain text in the diagonal form:



Rail Fence Transposition Cipher

Now, following the second step we get our cipher text.

Cipher Text= hlorsnelkiha

Columnar Transposition Cipher

- We first pick a keyword for our encryption.
- Write the plaintext out in a grid where the number of columns is the number of letters in the keyword.
- Title each column with the respective letter from the keyword.
- Take the letters in the keyword in alphabetical order, and read down the columns in this order.
- If a letter is repeated, we do the one that appears first, then the next and so on.

- let's encrypt the message "The tomato is a plant in the nightshade family" using the keyword *tomato*. We get the grid
- We have written the keyword above the grid of the plaintext, and also the numbers telling us which order to read the columns in. Notice that the first "O" is 3 and the second "O" is 4, and the same thing for the two "T"s.

T	O	M	A	T	O
5	3	2	1	6	4
T	H	E	T	O	M
A	T	O	I	S	A
P	L	A	N	T	I
N	T	H	E	N	I
G	H	T	S	H	A
D	E	F	A	M	I
L	Y	X	X	X	X

Decryption

Decrypt the ciphertext "ARESA SXOST HEYLO IIAIE XPENG DLLTA HTFAX TENHM WX" given the keyword *potato*

P	O	T	A	T	O
4	2	5	1	6	3

Now we start by filling in the columns in the order given by the alphabetical order of the keyword, starting with the column headed by "A". After the first column is entered we have the grid shown to the right.

P	O	T	A	T	O
4	2	5	1	6	3
			A		
			R		
			E		
			S		
			A		
			S		
			X		

P	O	T	A	T	O
4	2	5	1	6	3
	O		A		
	S		R		
	T		E		
	H		S		
	E		A		
	Y		S		
	L		X		

After inserting the second column.

P	O	T	A	T	O
4	2	5	1	6	3
	O		A		O
	S		R		I
	T		E		I
	H		S		A
	E		A		I
	Y		S		E
	L		X		X

After inserting the third column.

Now we read off the plaintext row at a time to get "potatoes are in the nightshade family as well"

Block (Single Columnar) Transposition Cipher

Step 1: First we write the message in the form of rows and columns, and read the message column by column.

Step 2: Given a keyword, which we will use to fix the number of rows.

Step 3: If any space is spared, it is filled with null or left blank or in by (_).

Step 4: The message is read in the order as specified by the keyword.

Double Columnar Transposition Cipher

- The main objective of using a Double Columnar Transposition Cipher is to encrypt the message twice.
- It makes use of the Single Columnar Transposition technique but uses two times. It can use the same or different secret keys.
- The output obtained from the first encryption will be the input to the second encryption.

Double Columnar Transposition Cipher

Step 1: First we write the message in the form of rows and columns, and read the message column by column.

Step 2: Given a keyword, which we will use to fix the number of rows.

Step 3: If any space is spared, it is filled with null or left blank or in by (_).

Double Columnar Transposition Cipher

Now applying keyword 2:

Given Text = GEEKSFORGEEKS

Keyword = NICK

Keyword 2= BOAT

N	I	C	K
1	2	3	4
E	E	K	G
O	F	R	S
E	E	K	G
_	_	_	S

Cipher Text = EOE_EFE_KRK_GSGS

Step 4: The message is read in the order in by the keyword

Double Columnar Transposition Cipher

Now apply step 3:

Given Text = GEEKSFORGEEKS

Keyword = NICK

Keyword 2= BOAT

B	O	A	T
2	3	1	4
G	E	E	K
S	F	O	R
G	E	E	K
S	_	_	_

Cipher Text = EOE_GSGSEFE_KRK_

Double Columnar Transposition Cipher

Step 5: Then the output from the first encryption is input to the second.

Step 6: Now the message is read in Technique in the order specified by the second keyword.

Given Text = GEEKSFORGEEKS

Keyword = NICK

Keyword 2= BOAT

B	O	A	T
1	2	3	4
E	G	E	K
O	S	F	R
E	G	E	K
_	S	_	_

EOE_GSGSEFE_KRK_

Cipher Text = EOE_GSGSEFE_KRK_

The Cipher Text is: **“EFEEOEGSGSKRK"**